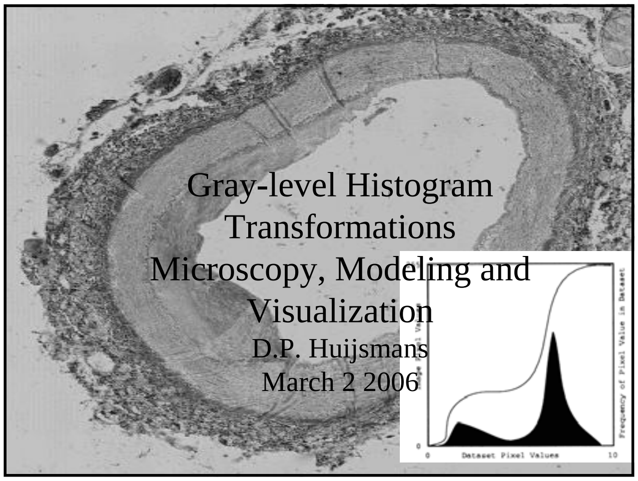


Gray-level Histogram Transformations

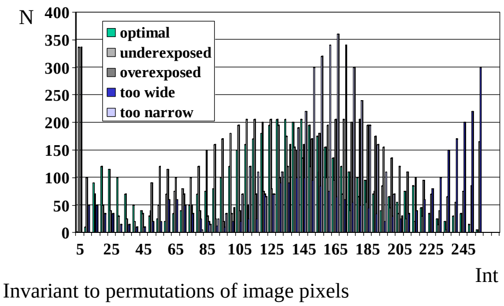
Microscopy, Modeling and Visualization

D.P. Huijsmans

March 2 2006

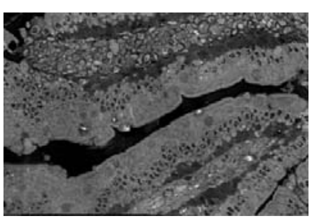
Gray-level Histograms

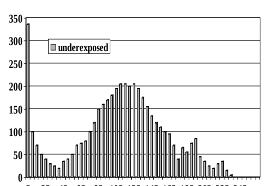
Image=Int(I,J) Histogram=N(Int)



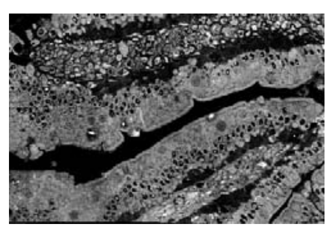
Invariant to permutations of image pixels

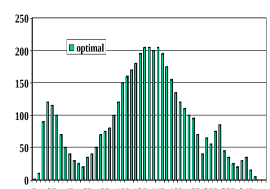
Histogram of Underexposed Image



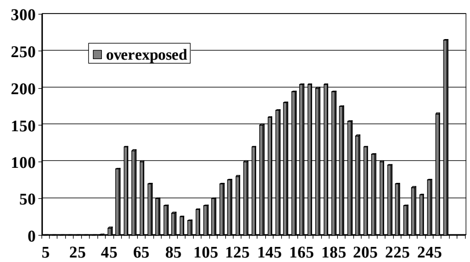


Histogram of Optimal Lighted Image

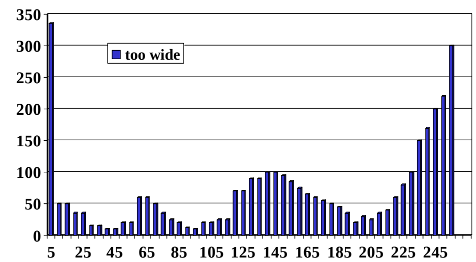




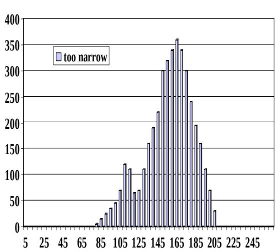
Histogram of Overexposed Image

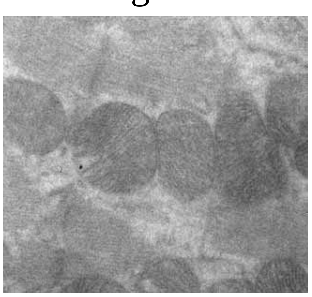


Histogram of Contrast Overstretched Image

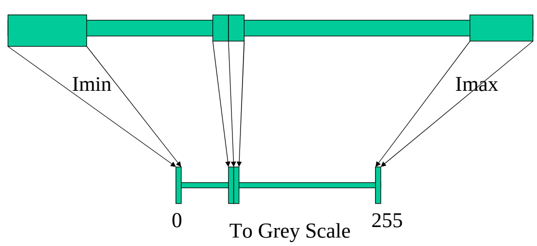


Histogram of Low Contrast Image





Dynamic Range Mapping



To Grey Scale
by
Shifting (Dark current level) and Scaling (Amplification)

Contrast Stretching

Using the Histogram

- On Input (Objective)

- On Display (Subjective)

Point Transformation: Iout = T (Iin) function of intensity only

Iout = (Iin-Imin)*[255/(Imax-Imin)]

Discrete number of in- and output values I in [0,255]

Table lookup: LUT hardware in video refresh cycle

Contrast Stretching

Using the Histogram

- On Input (Objective)

- On Display (Subjective)

shift scale

Point Transformation: Iout = T (Iin) function of intensity only

Iout = (Iin-Imin)*[255/(Imax-Imin)]

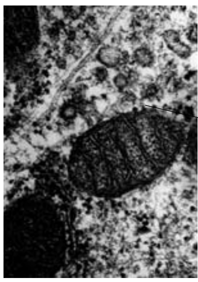
Discrete number of in- and output values I in [0,255]

Table lookup: LUT hardware in video refresh cycle

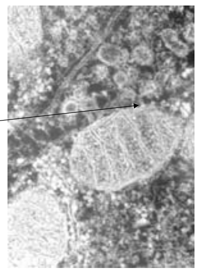
Discrete Intensity Transformation

- Also called histogram transformation
- For 8bit (byte) deep image 256 values
- Only 256 possible output values needed
- Lookup table (often hardware table)
- Input intensity used as table address
- Value at address in LUT is output value
- Fits in video read-out cycle

LUT Settings for Image Inversion



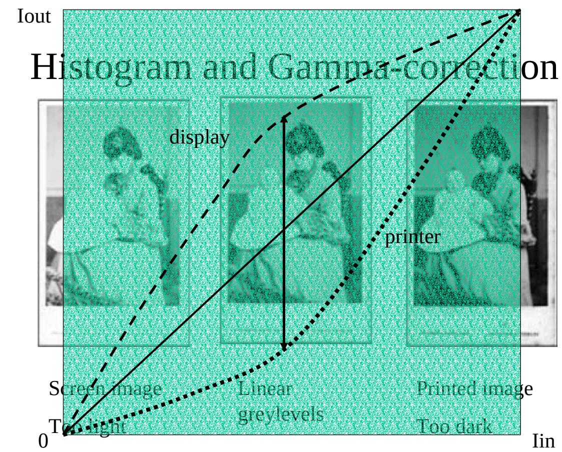
Iin	Iout=255-Iin
0	255
1	254
..	..
163	92
..	..
254	1
255	0



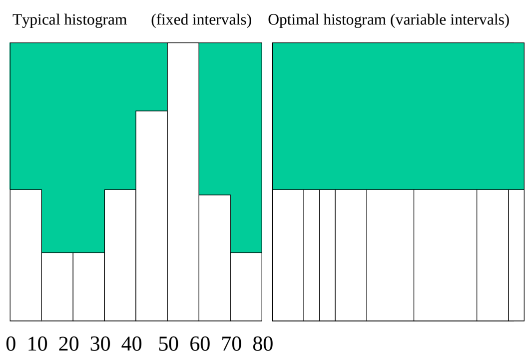
Histogram and Gamma-correction

Screen image	Linear greylevels	Printed image
Too light		Too dark

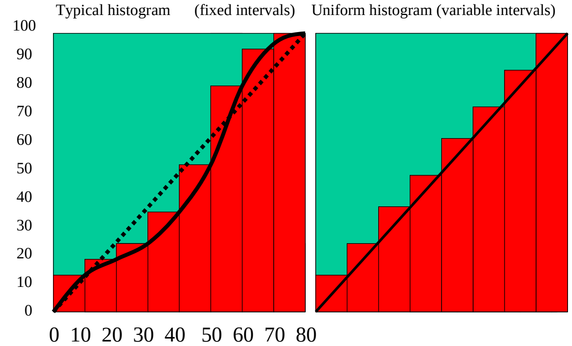
Histogram and Gamma-correction



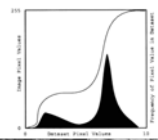
Grey-scale Usage



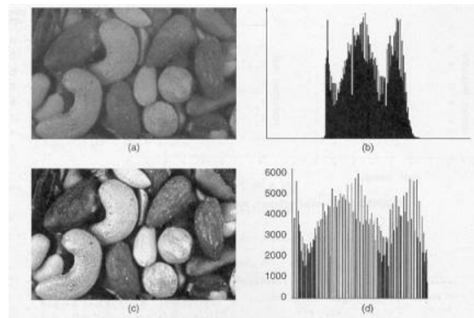
Cumulative Histogram



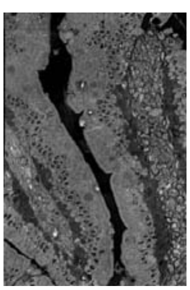
Approximating the ideal cum hist



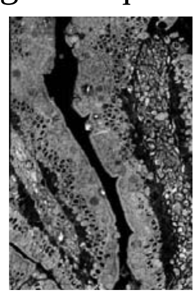
Intensity can only be relocated



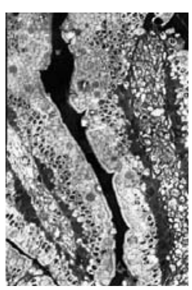
Histogram Equalisation



low contrast



Contrast stretched



Histogram equalized

Histogram Equalisation

Unsuited for binary-like grey-level images

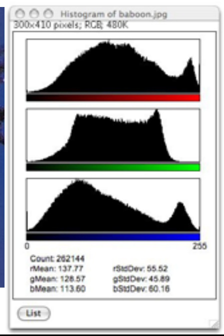
low contrast

Contrast stretched

High contrast

Histogram equalized

Color histograms: RGB or HSV channel wise transformations



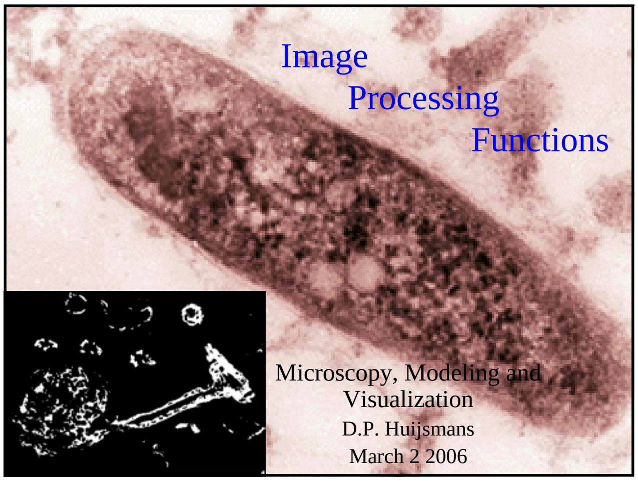


Image Types

Binary Image: 1 bit/pixel

Grayvalue Image: n bit/pixel (discretized continuous value)

Colour (multi-spectral) Image: n values/pixel

Labelled Image: n label values/pixel

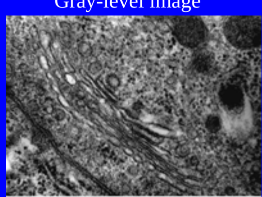
Image type and Interpolation

discretized continuous values can be interpolated;

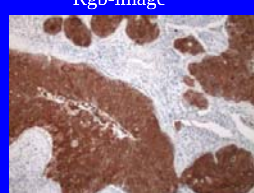
binary and labelled values cannot: choose a value

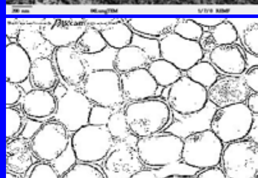
Image Types

Gray-level image



Rgb-image





Binary image

Color labeled image

Image Dimension

Although all images are flat 2D intensity distributions

their physical original dimensions determine

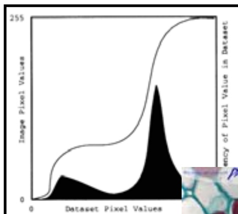
what can be done in image analysis

1D,t graphic image

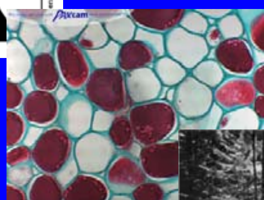
2D intersection image

3D projection image

Image Dimensions



graphic



Section 2D

Projection from 3D

Image File content

- Makes reconstruction of physical analog image possible
- Needed:
- - explicit intensities (regular samples) per channel
- - implicit coordinates:
- -- sampling resolution (dots per inch)
- -- number of rows
- -- number of columns

n Image File Formats (n>=120)

lossless, uncompressed:

essentially matrix values plus header (TIFF)

POSTSCRIPT: each byte value-> 2 printable ASCI chars

lossless, compressed:

runlength encoded (TIFF), GIF, PNG

lossy, compressed:

spatial frequency decomposition (JPEG, JPEG2000)

Image file conversions: n file formats->n.(n-1) conversion routines
with 1 intermediate format only 2n conversions needed

Computer Imagery

Output / Input	Analog Image	Digital Image	Non-Image data structure
Analog Image	Optics	Digitization /Sampling	Image Interpretation
Digital Image	Image Reconstruction	**Image Processing**	Image Analysis
Non-Image data structure	Drawing Painting	Computer Graphics	Rest of Computer Science

Image Processing

Transforms 1 or more input images into an output image

Intensity domain Transformations:

- point operations
- algebraic operations
- logic operations
- rank filter
- convolution filter
- geometrical transformation (later?)
- morphological transformation (later)

Spatial Frequency domain Transformations (later)

Image(s) to Image Transformations

Parameters:

Number of input matrices (mostly 1 or 2)

Pixel Intensity values

Pixel coordinate values (i,j or x,y)

$$\boxed{Int_{out}(i,j) = F(\,Int_{in1}(k,l),\ Int_{in2}(k,l), \ldots\ldots; Loc1(k,l), Loc2(k,l), \ldots)}$$

i,j,k,l in range [1,n]

Global recipe (algorithm) for calculating output pixel intensity and location from input intensities and/or locations

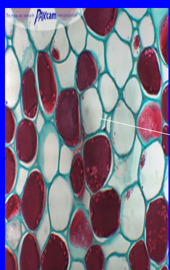
Image Processing Class Hierarchy

Nr Input Images / Nr pixels involved	1	>1
1	**Point operations**	**Algebraic Logic operations**
>1	**Neighbourhood filters**	

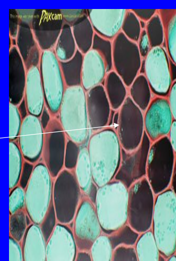
Point Operation Classes

Output Pixel Location / Output Pixel Intensity	Input Pixel Location	Shifted Input Pixel location
Input Pixel Intensity	**Image Copy**	**Integer Shifted/ Transposed/ Mirrored Image**
Transformed Input Pixel Intensity	**Intensity / Histogram Transformation**	Mixed Form

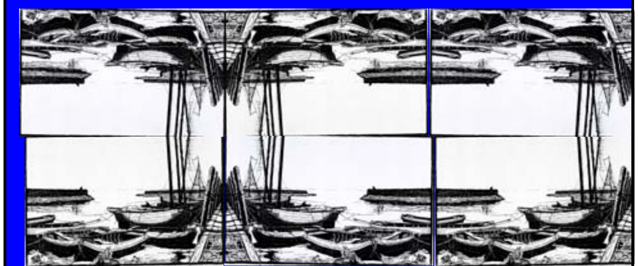
Intensity Transformation
Image Inversion



Iin	Iout=255-Iin
0	255
1	254
..	..
163	92
..	..
254	1
255	0



180 degree rotations by transpose and mirror



This is the way JPEG’s Discrete Cosine Transform sees the image

Algebraic Operations

$Int_{out}(i,j) = Int_{in1}(i,j)\ \{+,-,*,/\}\ Int_{in2}(i,j)$

Sum: ghost images

Difference: change detection (security), background removal

Multiply: ghost images

Divide: background equalization

Sum & Divide: noise reduction, moving object(s) removal

Sum & Multiply: weighted sum for dissolve and fade frames (alpha-channel)

Logic Operations (Bitwise)

$\boxed{Intout(i,j) = Intin1(i,j)\ \{AND,OR,XOR,\ NOT,=,<,<=\ldots.\}\ Intin2(i,j)}$

For Binary images (Bitmaps) or Bitplane by Bitplane

Spatial Set operations: Shape Arithmetic

Intersection, Union, difference, complement

ROI (crop), inlay, overlay,

A third bitmap may be used as a mask

Neighborhood Filter Classes

Location center pixel / Pixel values involved	Input center pixel location	Coordinate transformation
Only intensity values	**Rank Filter**	
Both intensity and location values	**Convolution Filter** **Morphological Transform**	**Geometrical Transformation with Intensity Interpolation**

Rank (or Percentile) Filter
statistical filter

Intensities in corresponding kxl neighborhood are **sorted**:

Certain percentile value transfered to output:

0% min filter, 50% **median filter**, 100% max filter

Min an max filter emphasize outliers;

Median filter is a very good noise reduction algorithm

Repeated application of the median filter produces an image with a painted appearance

Rank Filter example

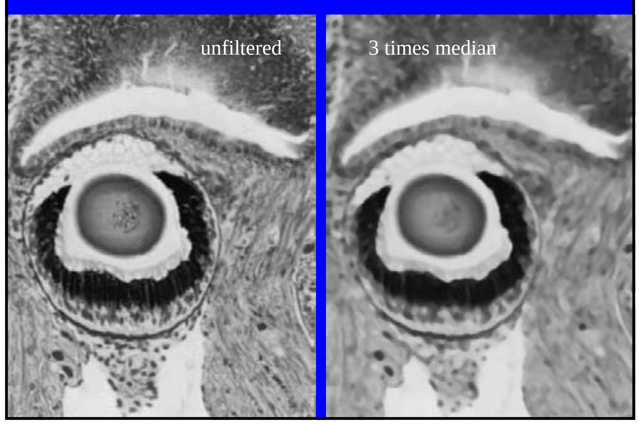
Around $Int_{In}(i,j)$=30

20	27	40
40	**30**	37
50	42	35

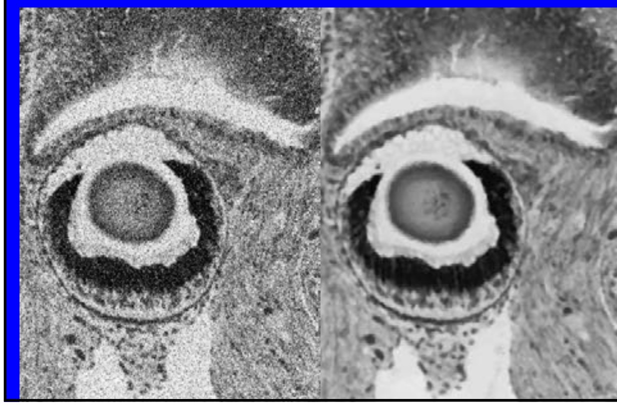
	0				50				100 %
rank	1	2	3	4	5	6	7	8	9
	20	27	30	35	37	40	40	42	50
	Min				Median				Max

Intout (i,j) = 20 Min
37 Med
50 Max

Rank Filter example



Noise removal with Median filter

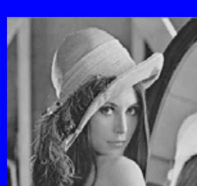


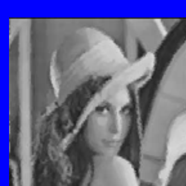
more complex statistical filters

- MAX-MIN filter (Range filter): largest local difference; sort of gradient image
- Kuwahara Filter: This filter is an edge-preserving filter. In each of four compass regions (NW,NE,SE,SW), the mean brightness and the variance are calculated. The output value of the center pixel in the window is the mean value of that compass region that showed the smallest variance.
  This filter is an edge-preserving filter, which smoothes the images without disturbing the sharpness and the position of edges.

Kuwahara Filter

nw	nw	n	ne	ne
nw	nw	n	ne	ne
w	w	c	e	e
sw	sw	s	se	se
sw	sw	s	se	se





Convolution or Weight Filter

Output value is a **weighted sum of intensity values** around the corresponding input pixel location (convolution)

The weights can depend on the location in the neighborhood

3x3 intensities around I5

I1	I2	I3
I4	I5	I6
I7	I8	I9

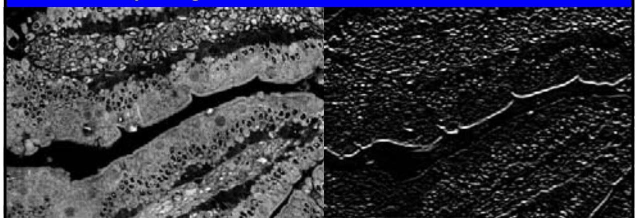
W1	W2	W3
W4	W5	W6
W7	W8	W9

3x3 Weight Coefficient Mask

O5 = I1*W1 + I2*W2 +…..+ I9*W9 (Dot Product)

Gradient Filter or Edge Detector

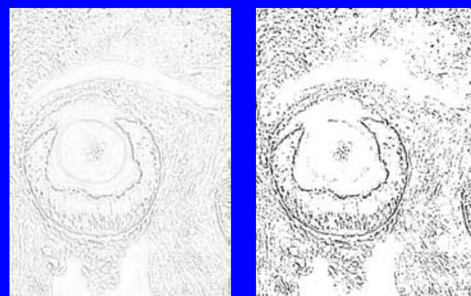
An important class of convolution filters are first- or second derivative filters that produce low values in uniform areas and high values in local areas with large intensity changes.



Horizontal edge magnitude

Thresholding Edge Maps

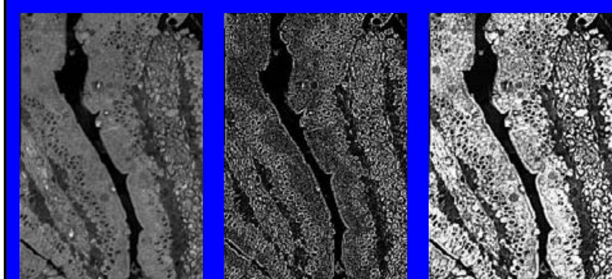
A drawing like binary image is obtained by a threshhold in the inverted gradient image



Gradients often coincide with object edges (but also with shadow edges!)

Image Sharpening

Adding a scaled gradient image to the original enhances local contrasts

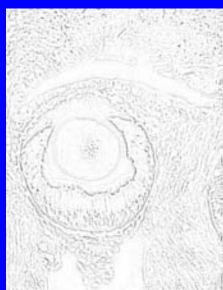


Some sharpening should be applied when the original image is smoothed by down sampling resolution or excessive compression

Unsharp masking with V-channel for color image

Positive and negative gradients

As in mountainous areas hills can go steep up or steep downwards

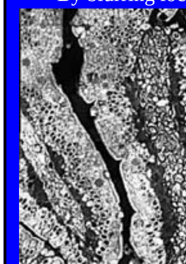


Absolute value of inverted gradient image

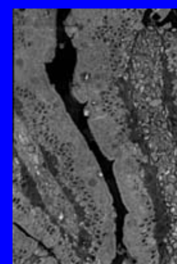
Signed gradient values around gray level halfway

Smoothing Filters

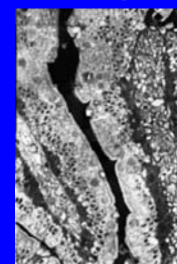
By blurring local contrast can be softened instead of sharpened



-1	0	1
-2	1	2
-1	0	1



0	0	0
0	1	0
0	0	0



1	2	1
2	4	2
1	2	1

*1/16